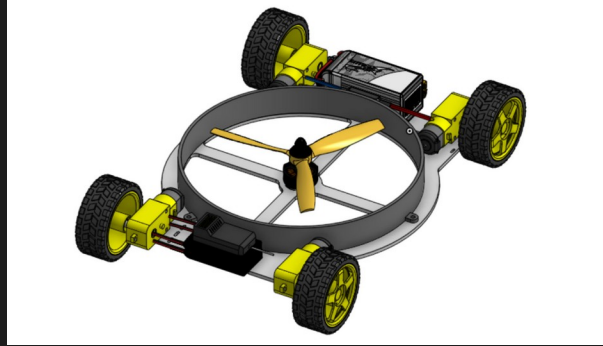


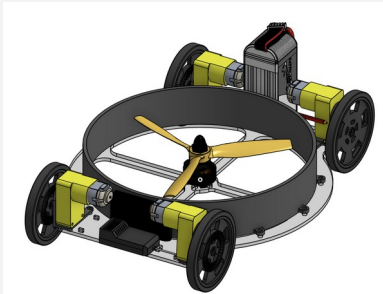
ENGR 7 FINAL



Project Overview

Objective

Design and build a compact, self-contained wall-climbing vehicle capable of completing both horizontal and vertical test courses while meeting cost, weight, and safety constraints.



Concept

The vehicle uses a fan-based adhesion system to create a pressure differential between the chassis and wall surface. This increases the effective normal force, allowing wheel friction to counteract gravity during vertical operation.

Engineering Approach

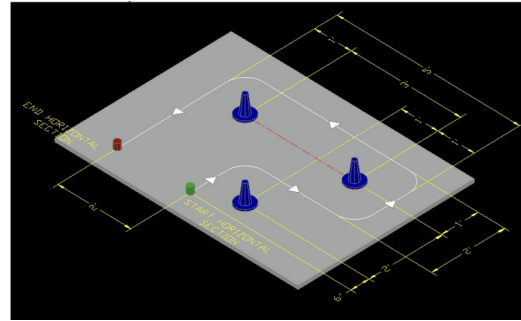
The design focuses on lightweight structural components, sufficient thrust generation, reliable wheel traction, and balanced weight distribution. CAD modeling, experimental testing, and iterative refinement were used to guide design decisions and evaluate performance.

Design Requirements

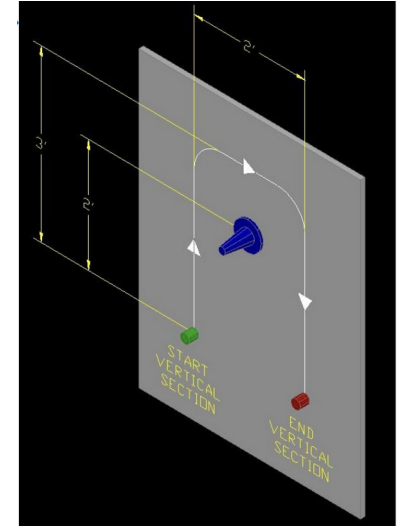
The wall-climbing vehicle was designed to meet the following requirements:

- Operate as a fully self-contained system with onboard power and control
- Navigate a short horizontal course and demonstrate vertical wall climbing
- Remain within the maximum weight limit specified by the project guidelines
- Stay under the allocated budget for primary mechanical and electrical components
- Utilize a safe and reliable adhesion and drive system suitable for indoor testing
- Be controllable using a simple remote or microcontroller-based control system

These requirements guided all design decisions, component selection, and testing throughout the development process.



Horizontal Course



Vertical Course

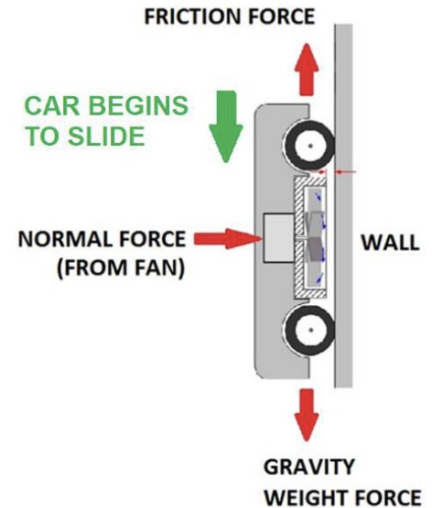
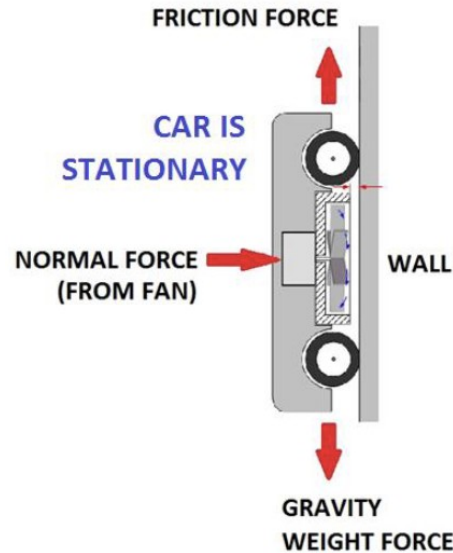
How the Vehicle Sticks to the Wall (Forces & Adhesion)

Wall Adhesion and Forces

During vertical operation, the vehicle must generate sufficient normal force to overcome its own weight and maintain traction on the wall surface. This is achieved using a fan-based adhesion system that creates a pressure differential between the underside of the chassis and the surrounding air.

By lowering the pressure beneath the vehicle, an effective downward force is produced, increasing the normal force between the wheels and the wall. The resulting friction force allows the vehicle to counteract gravity and remain attached during vertical motion.

The primary forces acting on the vehicle include gravitational force, fan-generated adhesion force, normal force, and frictional force. Proper balance between these forces is critical to ensure stable vertical climbing without slipping or detachment.



Friction Test (Static μ) — Results & Implication

Purpose

Estimate the coefficient of static friction (μ) between the drive wheels (rubber) and the test surface (wood) to evaluate available traction margin for vertical operation.

*Note on Test Vehicle
This friction test was performed using a **prototype vehicle with similar wheel material and loading**, not the final chassis. The result is used to estimate **traction capability and trends**, while final performance is verified during full vehicle testing.

Method (Ramp Test)

The vehicle was placed on an adjustable ramp. The angle was increased until initial sliding occurred (peak static friction). At the slip angle, the forces satisfy:

- $N = W \cos \theta$
- $F_{friction,max} = W \sin \theta$
- $\mu = \frac{F}{N} = \tan \theta$

Measured Data (Rubber on Wood)

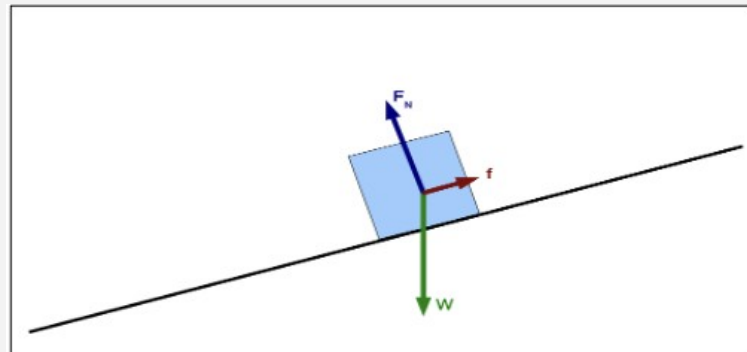
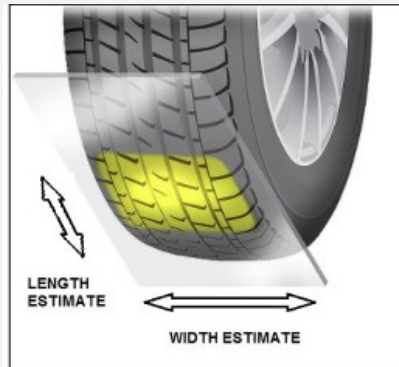
- $m = 0.357 \text{ kg}, \theta = 30.9^\circ \Rightarrow \mu \approx 0.598$

Key Observation

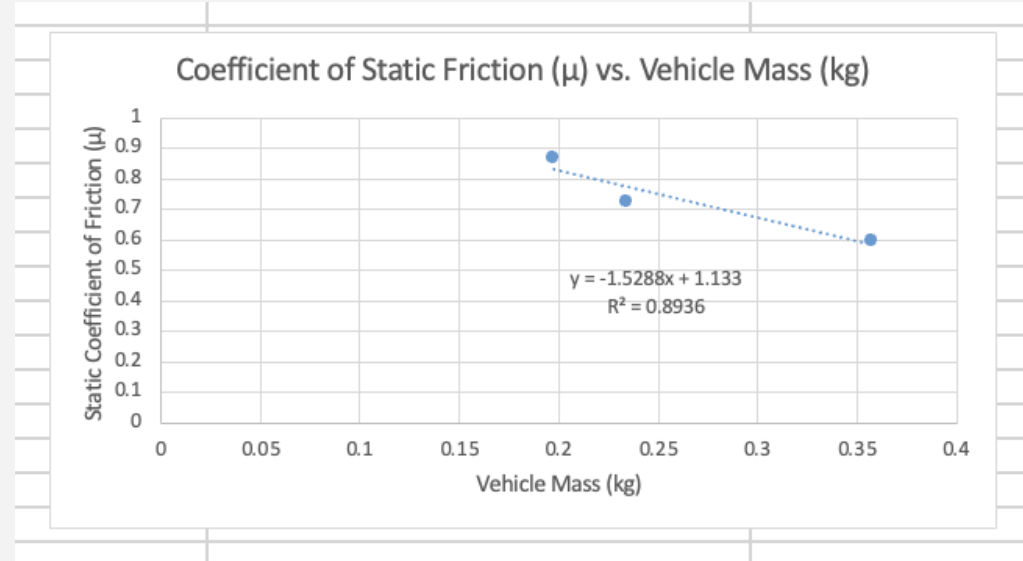
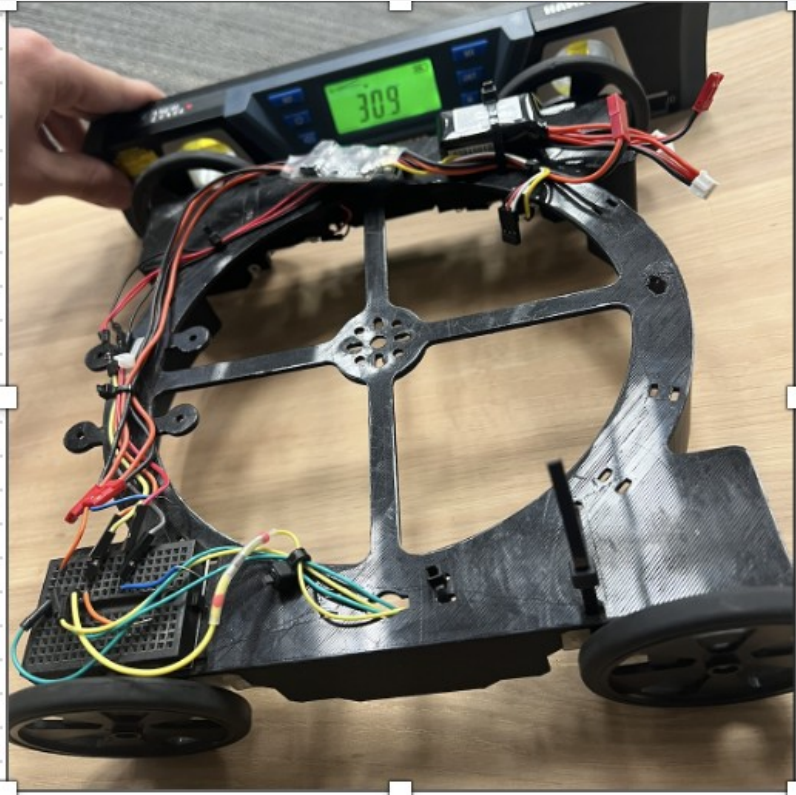
Measured μ decreased with increasing mass, indicating traction becomes more limiting as load increases. For design validation, the most conservative value is $\mu \approx 0.60$ (at the highest mass case).

Table: Static Friction Measurements and Calculated Coefficient of Friction for a Wheeled Vehicle on a Wood Ramp

VEHICLE MASS	CALCULATED VEHICLE WEIGHT (MASS*GRAVITY) WHERE GRAVITY = 9.8 M/S ²	MATERIAL (VEHICLE)	MATERIAL (RAMP)	APPROXIMATE SURFACE AREA (USE A ROUGH ESTIMATE OF THE TIRE CONTACT WIDTH*CONTACT LENGTH*NUMBER OF WHEELS)	MAXIMUM ANGLE (WHEN SLIDING BEGINS)
0.357	3.4986	RUBBER	WOOD	91	30.9
KG	N	N/A	N/A	SQ.MM	DEGREES
CALCULATED NORMAL FORCE AT MAX ANGLE (VEHICLE WEIGHT * COS(MAXIMUM ANGLE))		CALCULATED FRICTION FORCE AT MAX ANGLE (VEHICLE WEIGHT * SIN(MAXIMUM ANGLE))		COEFFICIENT OF FRICTION μ (FRICTION FORCE / NORMAL FORCE)	
3.002025879		1.796675424		0.598487654	
N		N		N/A	



Friction Test Results Summary



Thrust Test Results and Design Implications

Purpose

Evaluate the thrust output of the selected motor-propeller-battery configuration to determine whether sufficient normal force can be generated to support vertical wall climbing.

*Note on Test Vehicle

Method (Bench Thrust Data)

Thrust and efficiency data were obtained using publicly available thrust test results for a motor and propeller configuration closely matching the final design. The data includes thrust, electrical power, mechanical power, and system efficiency across a range of RPM values.

Key Observation

The measured thrust decreases with RPM, while overall system efficiency increases at lower operating speeds. For design evaluation, thrust values near the expected operating RPM were used to estimate available adhesion force.

Measured Thrust and Efficiency Data

TYTO TEST 1 (2300kV 6030 Carbon Nylon WITH 3S BATTERY)

MOTOR	BATTERY	PROPELLER	Time (s)	Throttle (%)	Rotation speed (rpm)	Thrust (N)	Torque (Nm)	Voltage (V)	Current (A)	Electrical power (W)	Mechanical power (W)	Motor & ESC efficiency (%)	Propeller efficiency (gf/W)	Propulsion system efficiency (gf/W)
EMAX RS11 2207 2300Kv	3S	6030 Carbon Nylon	1.236	1000	0	-0.001	0.000	11.141	0.015	0.165	0.000	0.000		
			19.868	1569	18350	0.461	0.040	10.969	9.548	104.738	76.270	72.820	6.041	4.399
			26.568	1550	18100	0.446	0.039	10.975	9.202	100.990	73.323	72.604	6.086	4.419
			33.293	1531	17807	0.435	0.037	10.980	8.869	97.390	69.627	71.494	6.241	4.462
			40.038	1512	17525	0.417	0.036	10.988	8.459	92.950	66.099	71.113	6.302	4.482
			46.777	1493	17195	0.401	0.035	10.995	8.092	88.967	63.070	70.891	6.363	4.511
			53.524	1474	16829	0.380	0.033	11.003	7.622	83.866	58.779	70.086	6.471	4.535
			60.295	1454	16510	0.360	0.032	11.011	7.183	79.094	54.632	69.072	6.586	4.549
			67.009	1435	16068	0.340	0.030	11.019	6.722	74.063	50.256	67.855	6.772	4.595
			73.752	1416	15706	0.326	0.028	11.025	6.347	69.975	46.751	66.811	6.969	4.656
			80.491	1397	15308	0.304	0.027	11.035	5.882	64.907	42.494	65.469	7.156	4.685
			87.268	1378	14787	0.282	0.025	11.044	5.350	59.083	38.067	64.430	7.418	4.780
			94.068	1359	14287	0.262	0.023	11.052	4.941	54.611	34.930	63.962	7.506	4.801
			100.890	1340	13755	0.238	0.021	11.061	4.429	48.990	30.537	62.333	7.804	4.865
			107.702	1321	13185	0.219	0.020	11.068	3.986	44.115	26.986	61.171	8.098	4.954
			114.501	1302	12537	0.197	0.018	11.076	3.492	38.677	23.094	59.711	8.519	5.087
			121.303	1283	11759	0.169	0.015	11.086	2.959	32.803	19.043	58.052	8.893	5.163
			128.089	1264	10988	0.146	0.013	11.094	2.481	27.529	15.503	56.315	9.405	5.296
			134.980	1244	10055	0.118	0.011	11.104	1.967	21.841	11.628	53.240	10.176	5.417
			141.897	1225	9097	0.096	0.009	11.111	1.543	17.147	8.773	51.162	10.980	5.618
			148.768	1206	8026	0.073	0.007	11.118	1.149	12.777	6.232	48.773	11.671	5.692

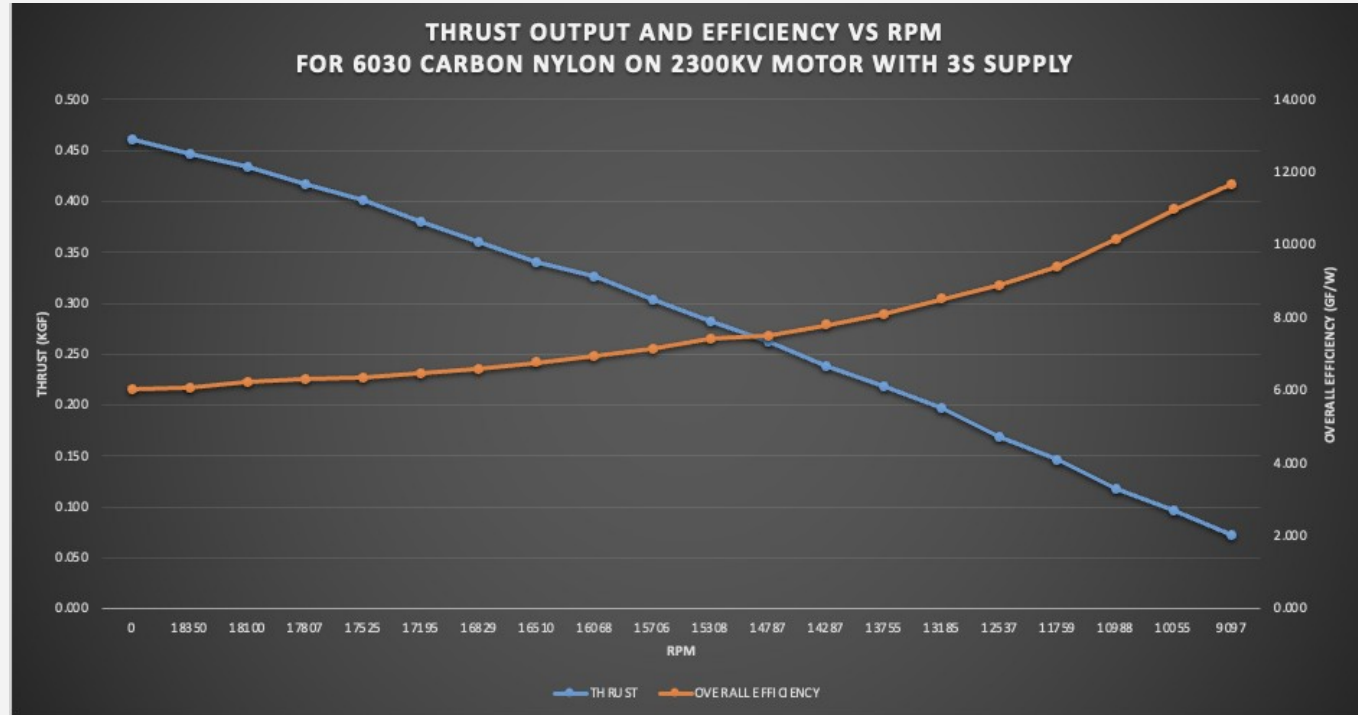
Table shows thrust output, electrical power, mechanical power, and system efficiency for a motor–propeller configuration similar to the final design.

*Note on Test Configuration

This thrust data was collected using a **similar motor and propeller model**, not the exact final hardware. Due to the close match in motor KV rating, propeller size, and battery voltage, the data provides a reasonable approximation for evaluating thrust capability and design feasibility. Final performance is verified during full vehicle testing.

Thrust Test Results Summary

- Thrust decreases with RPM, while overall efficiency improves at lower operating speeds.
- Operating RPM was selected to balance sufficient thrust and acceptable system efficiency.



Required Normal Force for Vertical Climbing

Design Requirement	Force Relationship	Calculated Requirement	Design Implication
For vertical wall climbing, the friction force generated at the wheels must be sufficient to counteract the vehicle weight.	The maximum available friction force is given by: $F_{\text{friction,max}} = \mu N$ To prevent slipping during vertical operation: $\mu N \geq W \quad \Rightarrow \quad N_{\text{required}} \geq \frac{W}{\mu}$	Using the most conservative measured friction coefficient: - Vehicle mass: $m = 0.357 \text{ kg}$ - Weight: $W = mg \approx 3.50 \text{ N}$ - Static friction coefficient: $\mu \approx 0.60$ $N_{\text{required}} \geq \frac{3.50}{0.60} \approx 5.8 \text{ N}$	This result indicates that the adhesion system must generate at least 5.8 N of normal force to maintain vertical traction. Thrust measurements from the selected motor-propeller configuration were evaluated to determine whether this requirement could be met.

Comparison of Required Normal Force and Available Thrust

- **Objective**

Compare the required normal force for vertical climbing with the available thrust generated by the propulsion system to evaluate overall feasibility.

- **Required Force (from Friction Analysis)**

From static friction testing, the conservative coefficient of friction was determined to be:

$$\mu \approx 0.60$$

For the vehicle mass $m = 0.357$ kg:

$$W = mg \approx 3.50 \text{ N}$$

$$N_{\text{required}} = \frac{W}{\mu} \approx \frac{3.50}{0.60} \approx 5.8 \text{ N}$$

- **Available Force (from Thrust Testing)**

Bench thrust data for a motor-propeller configuration similar to the final design shows that the propulsion system is capable of generating thrust values on the order of several newtons within the expected operating RPM range.

- **Feasibility Assessment**

Since the measured thrust exceeds the required normal force calculated from friction testing, the vehicle is theoretically capable of maintaining sufficient traction for vertical wall climbing. This comparison confirms that the selected propulsion and adhesion approach is feasible within the tested operating conditions.

Design Process — Chassis Development

Chassis Design Evolution

Initial Design (Instructor Baseline)

The initial chassis design provided a functional baseline geometry to support the propulsion system and major components. While structurally sound, the original layout was not optimized for weight reduction, component integration, or airflow performance required for vertical climbing.

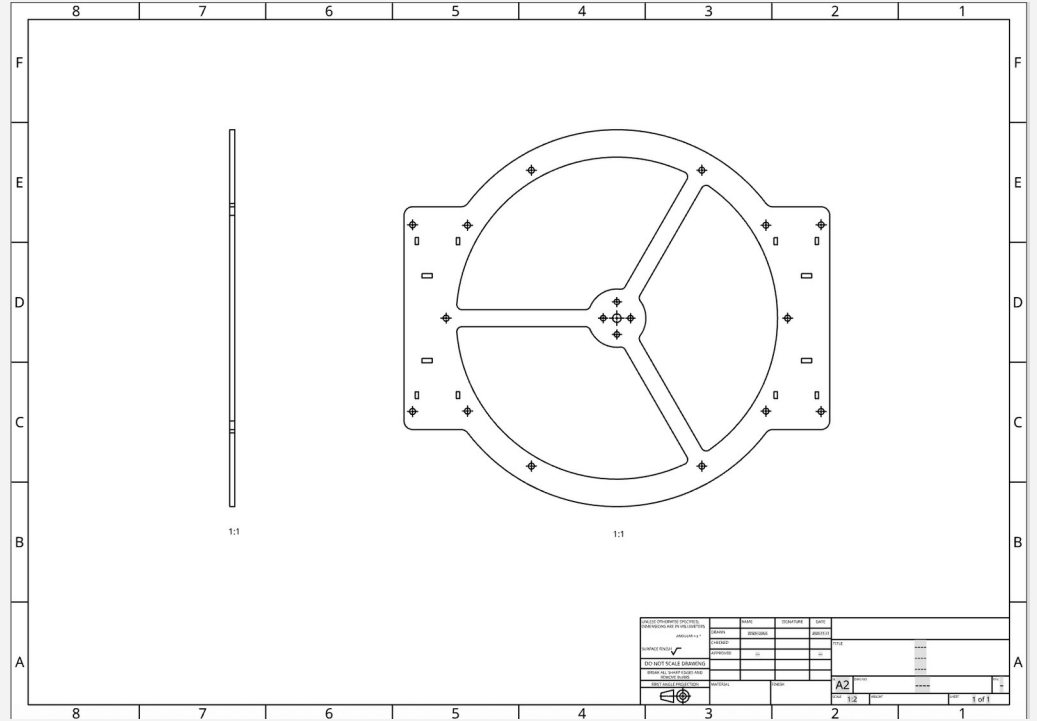
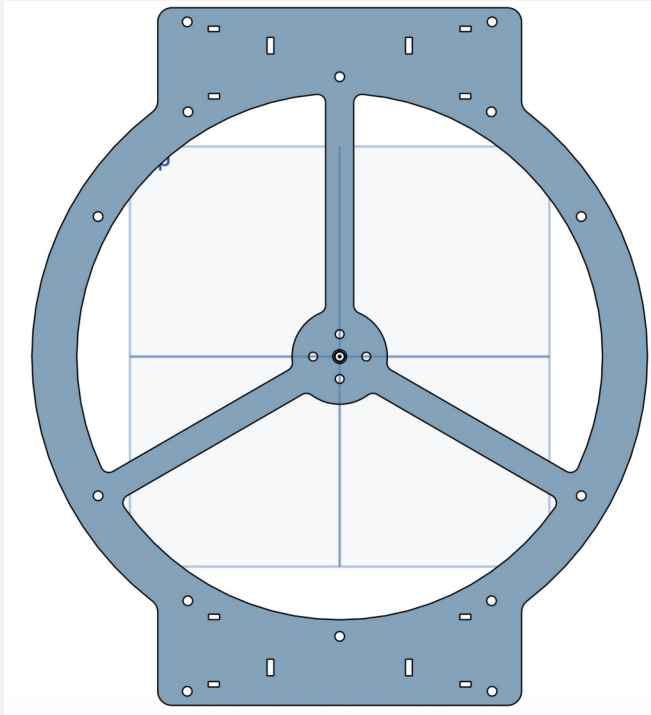
Revised Design (Final Iteration)

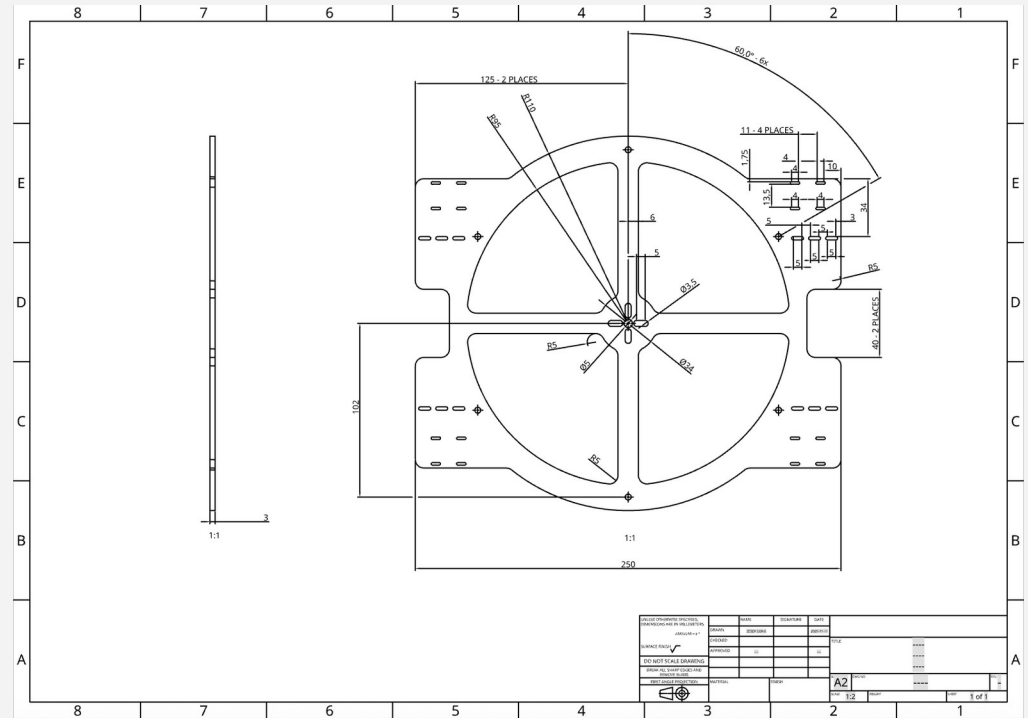
The chassis was redesigned to improve structural efficiency and reduce overall mass while maintaining rigidity. Material was removed from low-stress regions, mounting features were refined, and symmetry was improved to enhance balance and manufacturability.

Design Objectives

- Reduce chassis mass to increase available adhesion margin
- Maintain structural stiffness to prevent deformation near the fan opening
- Improve airflow path and clearance for the propulsion system
- Provide standardized mounting points for electronics and drivetrain components

Initial Design (Instructor Baseline)





Design Process — Fan Shroud Development

Shroud Design Evolution

Initial Concept

The initial shroud geometry was designed to surround the propeller and provide a basic duct for airflow containment. While sufficient for mounting and protection, the early design did not fully optimize airflow direction or pressure retention near the wall surface.

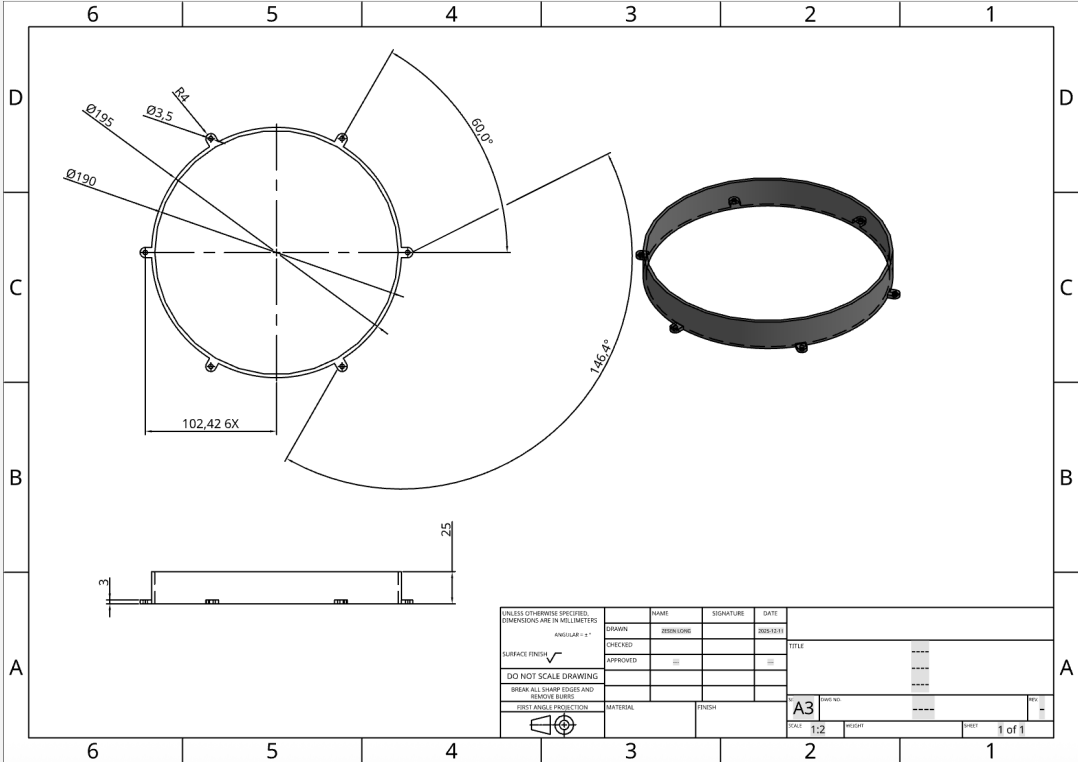
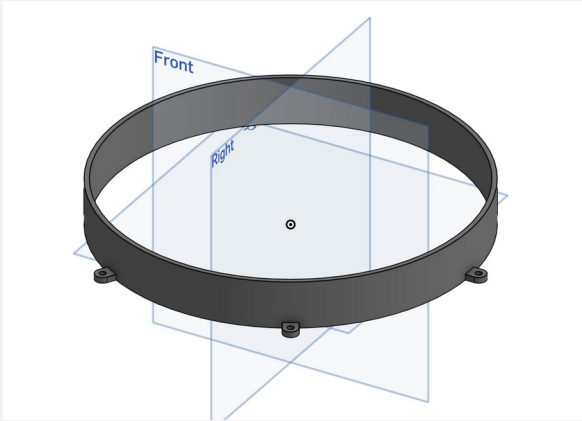
Revised Design (Final Iteration)

The shroud was refined to improve airflow control and structural integration with the chassis. The final design features a continuous circular wall with a uniform cross-section and multiple mounting tabs to ensure accurate alignment and rigidity during operation.

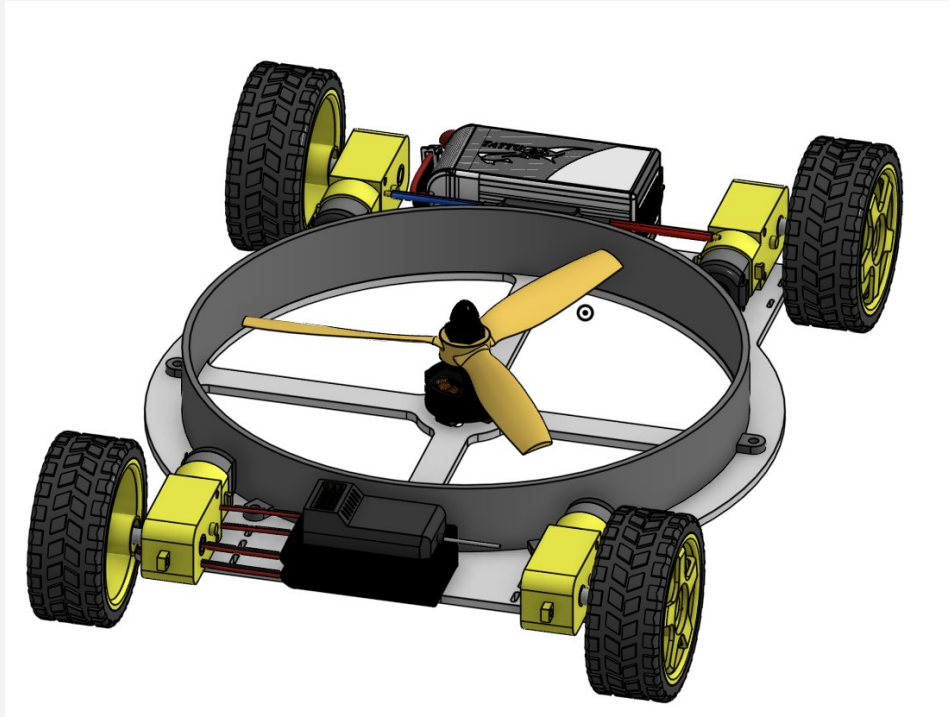
Design Objectives

- Contain and direct airflow to reduce losses at the propeller perimeter
- Improve pressure retention beneath the vehicle to increase effective adhesion force
- Maintain sufficient stiffness to prevent deformation under thrust loading
- Provide repeatable mounting and alignment relative to the chassis and propeller

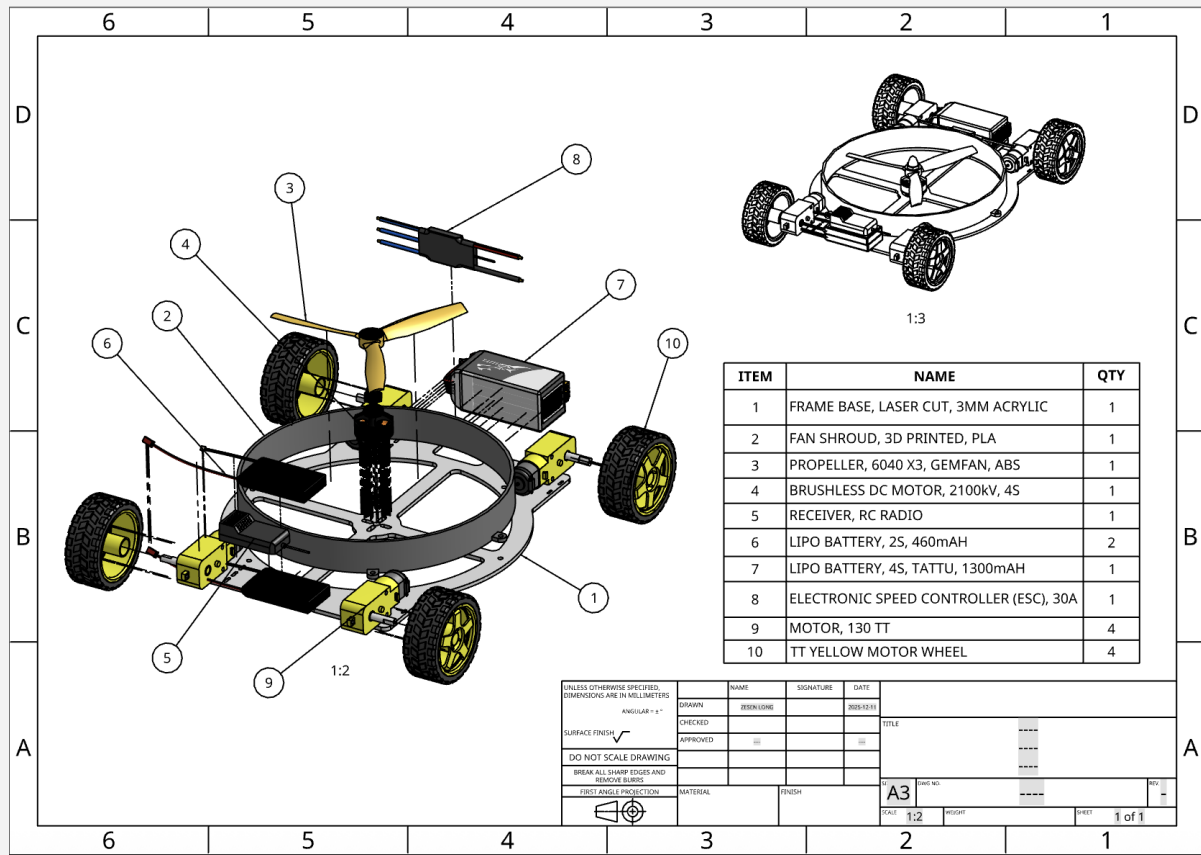
Revised Design (Final Iteration)



CAD Assembly Overview



Assembly Drawing and Bill of Materials

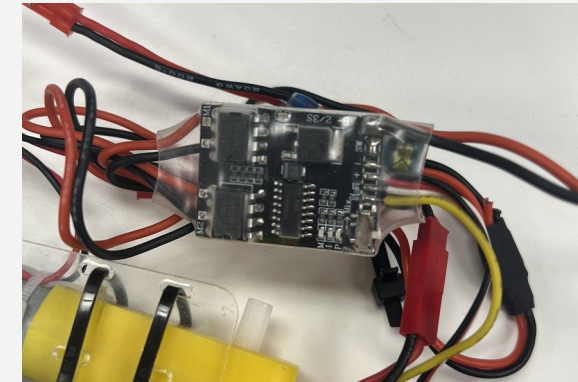


Final Bill of Materials (BOM) and Cost Breakdown

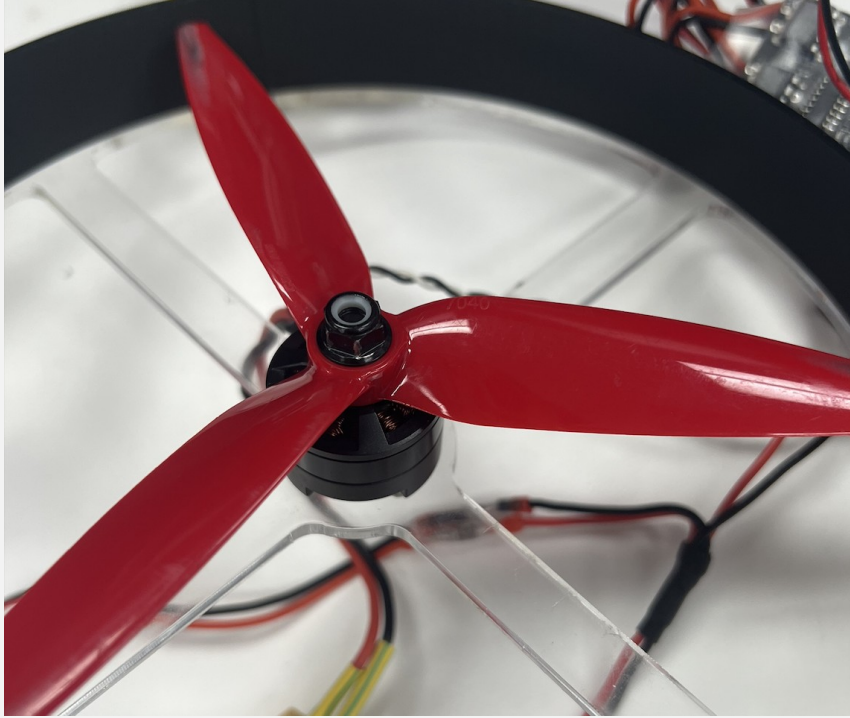
Includes motors, batteries, ESCs, receiver, propeller, wheels, and associated wiring

The components shown here represent the final hardware used in the assembled vehicle. While some parts differ slightly from the initially recommended options, all selected components are functionally equivalent and compatible with the system design.

These selections reflect real-world constraints such as availability, cost, and ease of integration, while maintaining performance requirements for propulsion, power delivery, and remote control.



Final Bill of Materials (BOM) and Cost Breakdown



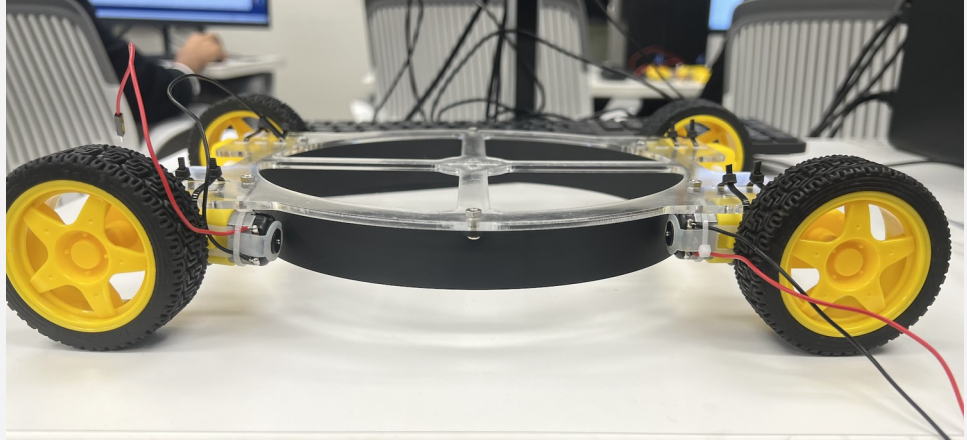
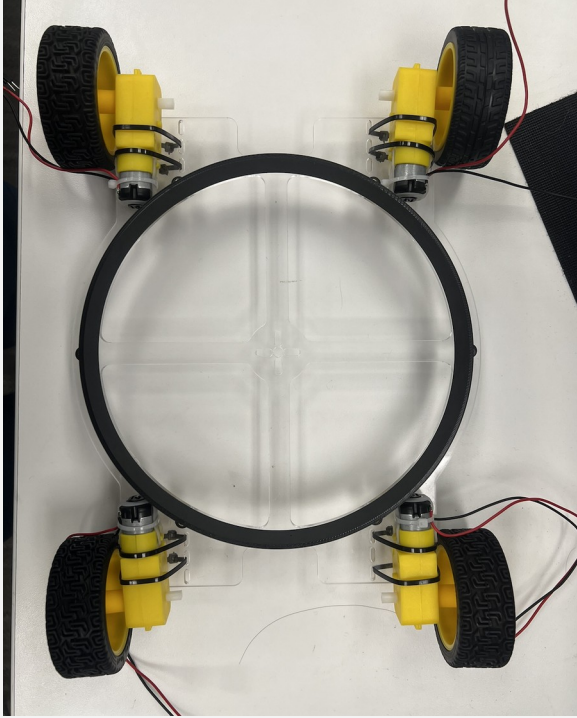
Final Bill of Materials (Actual Components Used)

Item	Component	Specification	Qty	Unit Price (USD)	Subtotal (USD)
1	DC Gear Motor + Wheel	TT Yellow Motor, 3–6V, with rubber wheel	4	\$3.75	\$15.00
2	Brushed Motor ESC	YoungRC Brushed ESC, 30A (controls 2 motors)	2	\$7.40	\$14.80
3	Brushless Motor (Fan)	2204 / 2100–2300KV class	1	\$10.00	\$10.00
4	Brushless ESC (Fan)	30A BLHeli ESC	1	\$10.00	\$10.00
5	Propeller	6040 / 6040x3 propeller	1	\$1.50	\$1.50
6	LiPo Battery (Drive)	2S 7.4V 450–460mAh	1	\$8.00	\$8.00
7	LiPo Battery (Fan)	4S 14.8V 1100–1300mAh	1	\$11.30	\$11.30
8	RC Receiver	FlySky FS-iA6B, 2.4GHz	1	\$23.80	\$23.80
9	Chassis Plate	Laser-cut acrylic, 3 mm	1	\$5.00	\$5.00
10	Fan Shroud	3D printed PLA	1	\$4.00	\$4.00
11	Fasteners	M3 screws, nuts, washers	—	\$3.00	\$3.00
12	Zip Ties	Nylon zip ties (motor mounting)	8	\$0.10	\$0.80
13	Wiring & Connectors	JST, XT60, servo cables	—	\$3.00	\$3.00

Total Project Hardware Cost = \$113.20

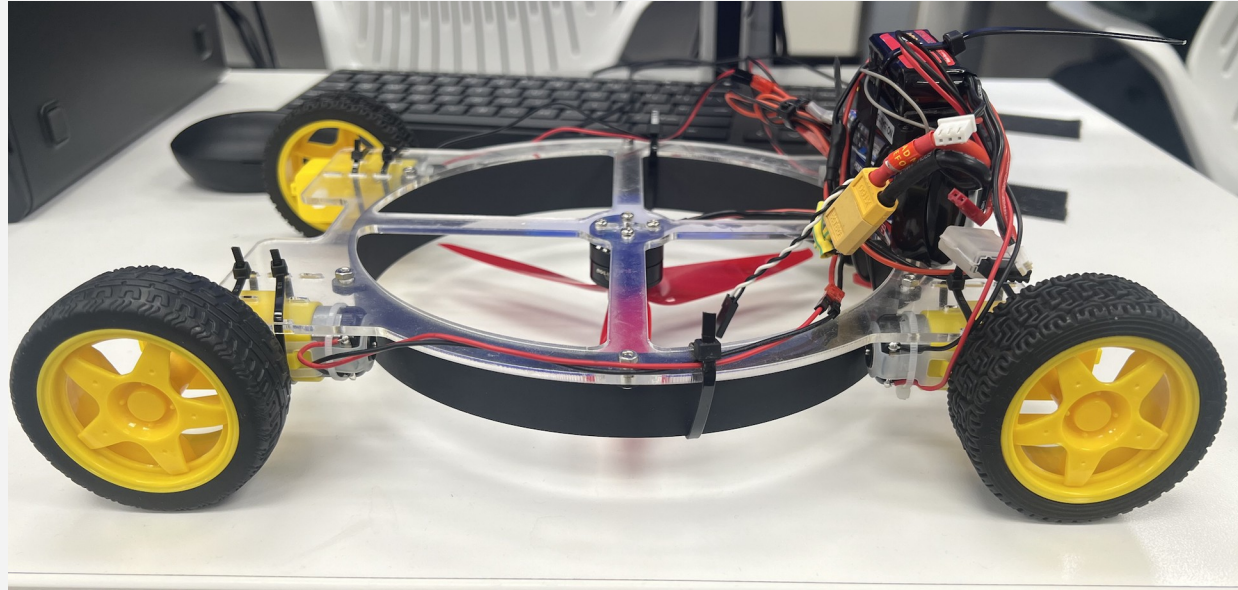
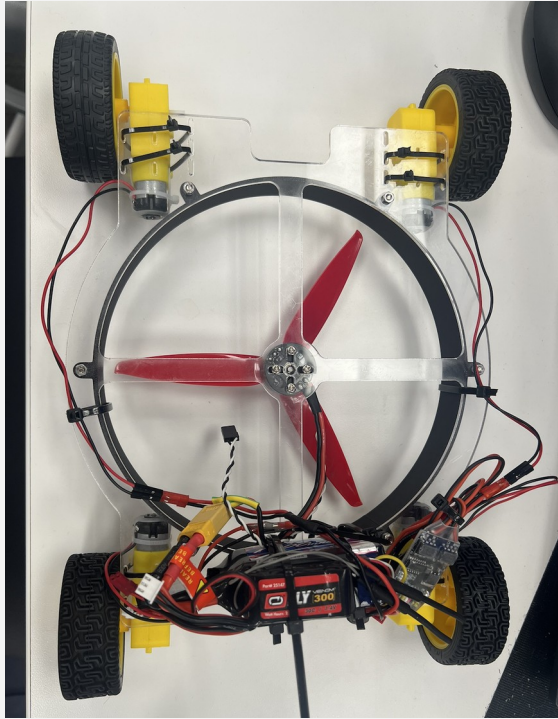
Several electronic components were reused from existing inventory. The effective cost of newly purchased parts remained within the \$100 project budget.

Motor Mounting and Shroud-Chassis Assembly

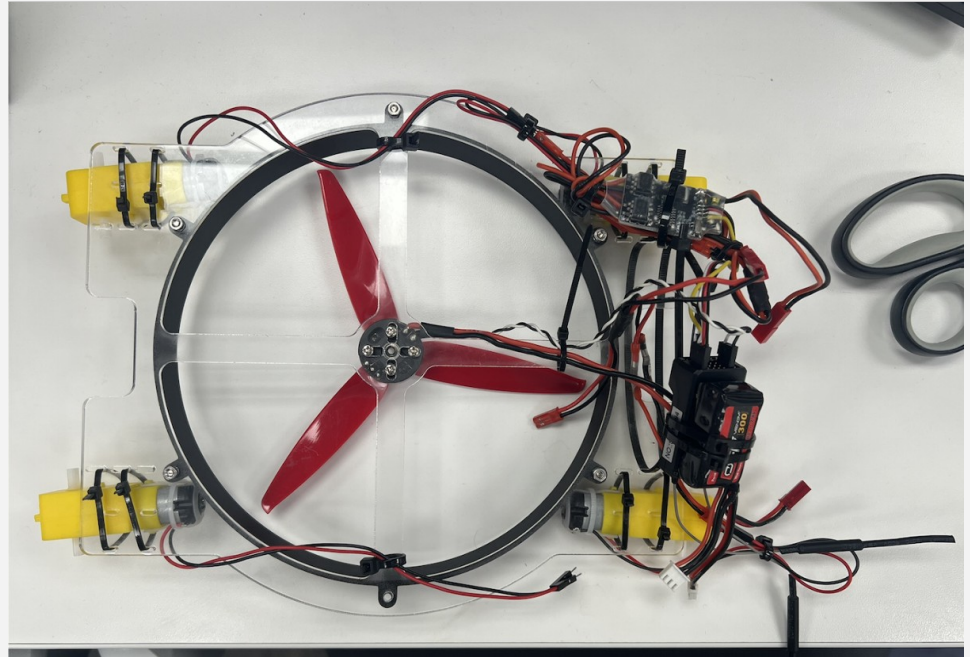
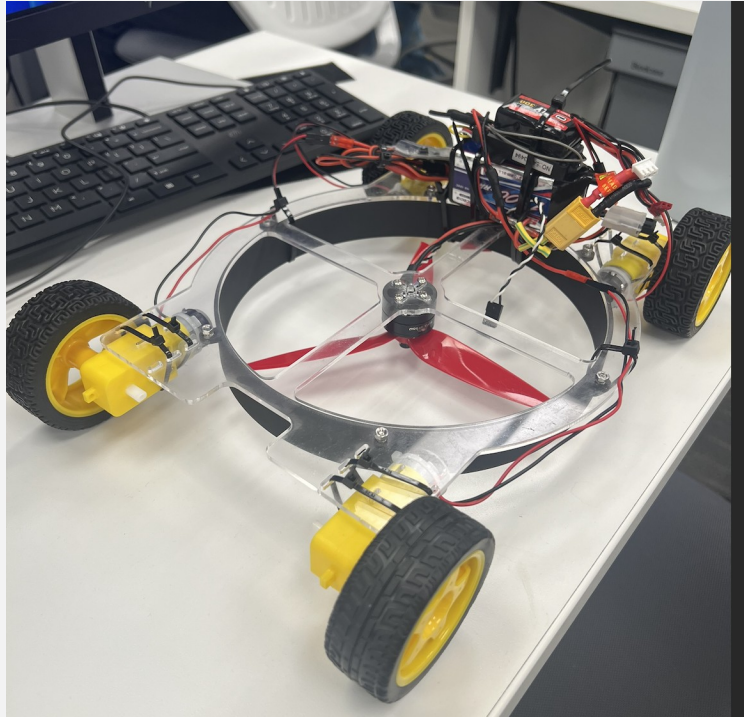


The motors were secured to the chassis using a total of **eight zip ties. The shroud was attached to the chassis using **six M3 screws**, providing a stable and rigid connection between the two components.*

Final Assembly



Final Assembly



Final Assembly and Zip Tie Mounting Iteration

The images above document the final assembled prototype of the vehicle and the evolution of the motor mounting method.

The first three images show the **initial zip tie mounting configuration**, which was later identified as suboptimal during testing. In this configuration, the motors were not sufficiently constrained, leading to slight misalignment and reduced structural stability under load.

The **final image shows the revised zip tie mounting method**, which was implemented after testing. In this updated configuration, the zip ties were repositioned and tightened to better secure each motor against the chassis, significantly improving alignment, rigidity, and overall drivetrain stability.

This final mounting method was used for all subsequent testing and represents the **final validated assembly configuration** of the vehicle.

Successful Vehicle Testing Results



Summary & Future Improvements

Project Performance

- Final cost remained within the \$100 budget limit
- Vehicle testing was successful and met all functional requirements
- Final vehicle weight was **630 g**, well below the maximum allowed limit

Design Limitations

- Fan shroud height was **25 mm**, below the required **38 mm**, which may reduce airflow efficiency
- Motor mounting holes were placed too close to the chassis edge, causing wheel interference and increased friction
- Motors were secured using zip ties, limiting alignment accuracy

Proposed Improvements

- Redesign the fan shroud to meet the **38 mm minimum height requirement**
- Increase spacing in motor mounting locations to prevent tire–chassis contact
- Design dedicated motor mounts for improved stability and alignment
- Add a battery mounting compartment for cleaner wiring and easier battery replacement

Conclusion

This project successfully demonstrated the complete engineering process from design and analysis to fabrication, assembly, and testing. The final vehicle met all budget and performance requirements, remained under the weight limit, and operated reliably during testing.

Through this project, key trade-offs between mechanical design, manufacturability, and real-world performance were identified. The results highlight the importance of dimensional planning, component integration, and iterative design refinement in engineering systems. Overall, the project achieved its intended goals and provided valuable hands-on experience in mechanical design, system integration, and testing validation.

Acknowledgements

I would like to sincerely thank everyone who supported and contributed to the successful completion of this final project.

First, I would like to thank **Professor Matthew Wolken** for his guidance throughout the course, including design feedback, technical instruction, material recommendations, and overall project direction. His support and clear expectations were essential to the completion of this project.

I would also like to thank **Professor Brett McKim** for his instruction in design model making and for sharing valuable insights related to fabrication and prototyping methods.

Special thanks to **Marlowe Patterson**, Laboratory Technician at **Irvine Valley College – School of Integrated Design, Engineering, and Automation (IDEA)**, for providing assistance with **3D printing, laser cutting, and lab equipment**, as well as for maintaining a safe and supportive prototyping environment.

Additionally, I would like to acknowledge the **IVC IDEA Department** for providing access to fabrication facilities, tools, materials, and workspace that made rapid prototyping and testing possible.

Finally, I appreciate the support of my classmates and the learning environment created through this course, which encouraged hands-on experimentation, iteration, and practical engineering problem-solving.

END

THANK YOU